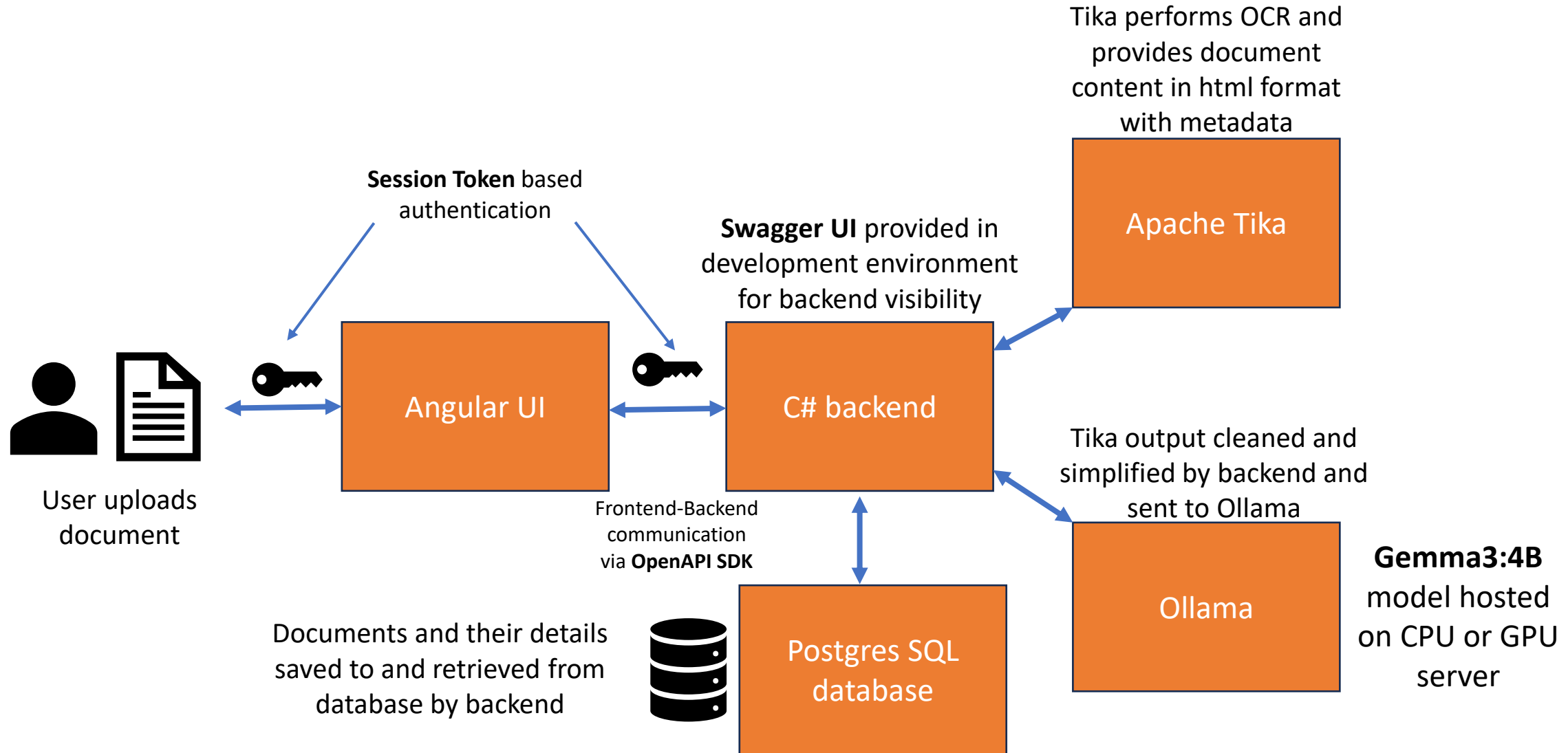


Architecture



System Description

- Session tokens for both Frontend and Backend expire in 1 hour if not logged out
- Uploaded documents can be PDFs or images
- Swagger UI available only in development environment
- Backend, Frontend, Postgres and Tika services are deployed using docker compose file, both in development and production environment (separate docker compose files for both environments)
- Only backend and frontend are accessible externally in both environments. Other services are accessible only internally by backend via Docker network
- Documents stored in dedicated Docker volume. Details of documents and buildings, with links to documents, stored in SQL database
- Ollama (with LLM) container deployed separately in a different docker compose file, to production server, and is externally accessible. Separate docker compose files for CPU-based and GPU-based servers
- Github workflows in place:
 - Deploying containers to production server on push to main
 - Deploying Ollama container on changes in Ollama configuration pushed to main
 - Backend and frontend linting checks on push to main or PR to main
 - OpenAPI SDK generation on push to main or PR to main
- Linting checks and OpenAPI SDK generation workflows need to be successful for PRs to be merged into main, unless overridden

Points to Note

- Apache Tika's raw output is too noisy for LLMs in Ollama, also increasing token count. Tika output must be converted to clean, flat, text
- Gemma3:4B model provided best trade-off between runtime and accuracy for both CPU-based and GPU-based hosting servers, among all major multilingual models available in Ollama repository, till date
- Ollama LLM model configurations can be set through environment variables in docker compose file for Ollama